Filip Farkašovský

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Profile

Mathematics and Economics graduate with a strong background in **statistics, machine learning, and mathematical modelling**. Experienced in developing **scientific computing software** and implementing numerical algorithms using Python, C++, and C#.

Technical Skills

Languages	:	Python, R, C#, JavaScript, MATLAB, C
Data Science	:	scikit-learn, Pandas, PyTorch, TensorFlow
DevOps	:	Docker, Kubernetes, Terraform, GitLab CI/CD, Nginx, uWSGI
Databases	:	SQL, PostgreSQL
Dev Tools	:	Git, Linux (Ubuntu, Fedora), Jira, Postman

Experience

Scientist Intern Jul 2026 – Present
Thermofisher Scientific *Brno, Czech republic*

- Research and development of numerical methods for electron diffraction simulation and pattern indexing
- Implementing high-performance scientific computing algorithms using C# and GPU acceleration

QA Engineer Intern Jul 2025 – Sep 2025
GEN(former Avast) *Brno, Czech republic*

- Supported automation testing to streamline QA processes
- API testing with Postman and reported issues with clear documentation
- Contributed to sprint planning, daily stand-ups, and regression testing
- Used Jira for bug tracking and sprint management in Scrum workflows

Education

M.Sc. Artificial Intelligence and Data Processing Sep 2026 – Present
Faculty of Informatics, Masaryk University

M.Sc. Mathematical and Statistical Methods in Economics Sep 2026 – Present
Faculty of Economics and Administration, Masaryk University

B.Sc. Mathematics and Economics Sep 2023 – Jun 2026
Faculty of Science, Masaryk University

- Ranked in the top 10% of students

Erasmus Exchange Programme Oct 2024 – Jan 2025
University of Murcia, Spain

Projects

Cloud Infrastructure & CI–CD

- Designed and deployed a scalable web architecture using Flask and PostgreSQL
- Containerized applications with Docker and orchestrated deployments using Kubernetes
- Implemented Infrastructure as Code (IaC) using Terraform
- Built GitLab CI/CD pipelines for automated testing and GitOps-based deployments

Neural Network from Scratch

- Implemented a feed-forward neural network from scratch in C++ without high-level numerical libraries
- Developed backpropagation and gradient-based optimization algorithms for network training
- Trained and evaluated the network on the Fashion-MNIST dataset